The equations we consider together in such settings make a meaningful situation and are known as simultaneous linear equations.

Real life Situation to understand the simultaneous linear equations

Consider the situation, Anitha bought two erasers and a pencil for ₹10. She does not know the individual cost of each. We shall form an equation by considering the cost of eraser as 'x' and that of pencil as 'y'.

That is $2x + y = 10$ ... (1)

Now, Anitha wants to know the individual cost of an eraser and a pencil. She tries to solve the first equation, assuming various values of x and y.

2 × cost of eraser + 1 × cost of pencil = 10

$2(1)+8 = 10$

$2(1.5)+7 = 10$

$2(2)+6 = 10$

$2(2.5)+5 = 10$

$2(3)+4 = 10$

⋮ ⋮

Points to be plotted :

x	1	1.5	2	2.5	3	...
y	8	7	6	5	4	...

She gets infinite number of answers. So she tries to find the cost with the second equation.

Again, Anitha needs some more pencils and erasers. This time, she bought 3 erasers and 4 pencils and the shopkeeper received ₹30 as the total cost from her. We shall form an equation like the previous one.

The equation is $3x + 4y = 30$ ...(2)

Even then she arrives at an infinite number of answers.

3 × cost of eraser + 4 × cost of pencil = 30

$3(2)+4(6) = 30$

$3(4)+4(4.5) = 30$

$3(6)+4(3) = 30$

$3(8)+4(1.5) = 30$

⋮ ⋮

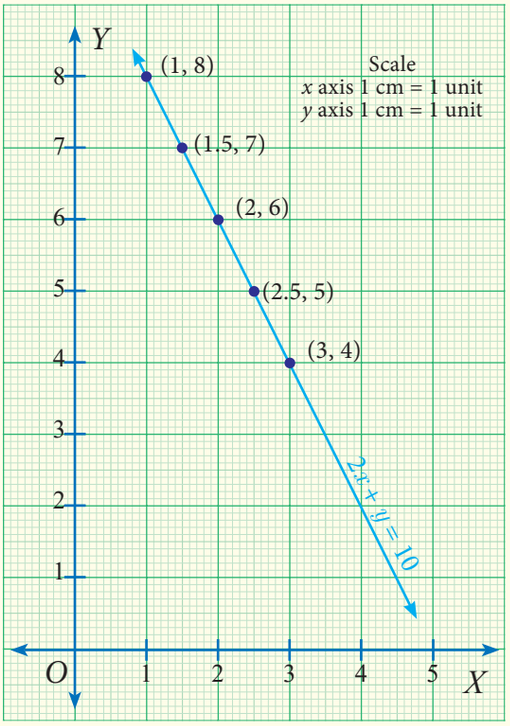


Fig. 3.19

3 Algebra_Term1.indd 119 02/12/2022 21:40:10